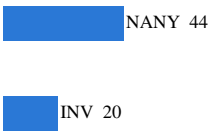
TritVerilog

```
generate
 for (i=0; i<N; i=i+1)
 begin : digit
  full_add fa (
   .a(a[i]), .b(b[i]),
   .cin(c[i]),
   .sum(sum[i]),
   .cout(c[i+1]));
 end
endgenerate
```

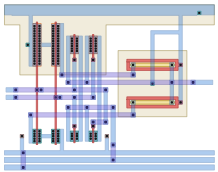
adder #(N = 4): 17 lines
+ full_add, half_add modules

Synthesis



64 cells, depth 27
equivalence: exhaustive

Standard cells



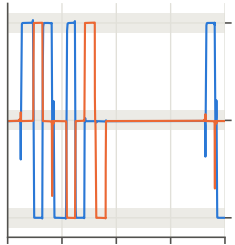
9 cells from taped-out circuits
132 µm tall, three rails

Place & route



1692 × 1590 µm, M1 + M2
0 DRC markers, LVS match

Post-layout SPICE



sum[3] (blue), cout (orange), 0–10 µs
extracted devices + wire C